ENGS 104 Overview - Fall 2026

Calendar

Instructors: Dr. Bryce L. Ferguson and Dr. Armin Fügenschuh

Part I: Linear Programming

Date	Topic	Assignment
Tu (9/15)	Linear Programs, Standard Simplex Method	
Th (9/17)	Duality Theory	HW1 assigned
F(x) (9/18)	Dual Simplex Method	
Tu (9/22)	Game Theory (Zero-Sum Matrix Games)	
Th (9/24)	Revised Simplex Method (Primal and Dual)	
F(x) (9/25)	Mixed-Integer Programming, Branch-and-Bound	
Tu (9/29)	Cutting Plane Methods I	
Th (10/1)	Cutting Plane Methods II	HW2 assigned
F(x) (10/2)	Decomposition Methods I (Lagrange)	HW1 due
Tu (10/6)	Decomposition Methods II (Benders, Dantzig-Wolfe)	
Th (10/8)	Stochastic Optimization	
F(x) (10/9)	TA Session - Multi-Variable Calculus Review	

Part II: Nonlinear Programming and Convex Optimization

Date	Topic	Assignment
Tu (10/13)	Nonlinear Optimization Problems	
Th (10/15)	Convex Functions and Convex Sets	HW3 assigned
F(x) (10/16)	TA Session - CVXPY	HW2 due
Tu (10/20)	Classes of Convex Programs	
Th (10/22)	Robust Optimization and Conic Programming	Project Proposal due
F(x) (10/23)	Project Proposal Discussions	
Tu (10/27)	Duality I	
Th (10/29)	Duality II	HW4 assigned
F(x) (10/30)	TA Session - Solver Tools	HW3 due
Tu (11/3)	Algorithms & Solvers I	
Th (11/5)	Algorithms & Solvers II	
F(x) (11/6)	Distributed Optimization	
Tu (10/10)	Applications: Estimation, Regression, and SVM	
Th (10/12)	Applications: Optimal Control	
F(x) (10/13)	Applications: Network Routing and Population Games	HW4 due
Tu (10/17)	Final Presentations	
Th (10/19)	Final Presentations	
F(x) (10/20)	Final Presentations	Project Report due

Lecture Outlines

Part I: Linear Programming

Linear Programs, Standard Simplex Method:
Modeling examples, standard forms, feasible regions and polyhedra, basic feasible solutions, geometry of linear programming, optimality at extreme points, simplex tableaux, pivoting, reduced costs, degeneracy, cycling, Bland's rule, unboundedness and infeasibility

Duality Theory:
Primal and dual linear programs, weak and strong duality, complementary slackness, economic interpretation of dual variables, shadow prices, sensitivity analysis, certificates of optimality and infeasibility

Dual Simplex Method:
Primal feasibility vs dual feasibility, motivation for the dual simplex method, dual feasible bases, pivot rules, maintaining dual feasibility while restoring primal feasibility, applications in reoptimization, post-optimality analysis, and branch-and-bound

Game Theory (Zero-Sum Matrix Games):
Two-player zero-sum games, pure and mixed strategies, payoff matrices, minimax theorem, formulation as primal-dual linear programs, value of a game, optimal strategies, examples from competitive decision making

Revised Simplex Method (Primal and Dual):
Computational limitations of tableau methods, basis matrices, reduced costs, pricing, ratio tests, updating the basis inverse, primal revised simplex, dual revised simplex, sparse linear algebra, numerical considerations, connection to modern LP solvers

Mixed-Integer Programming, Branch-and-Bound:
Integer and mixed-integer linear programs, modeling with binary variables, logical constraints, fixed charges, disjunctive constraints, big-M formulations, LP relaxations, integrality gaps, branch-and-bound trees, bounds, fathoming, incumbent solutions, branching rules, solver behavior

Cutting Plane Methods I:
Motivation for cutting planes, strengthening LP relaxations, valid inequalities, separating fractional LP solutions, Chvatal–Gomory cuts, Gomory fractional cuts, geometric interpretation, finite convergence ideas, examples for pure integer programs

Cutting Plane Methods II:
Mixed-integer cuts, split cuts, cover inequalities, clique inequalities, flow-cover cuts, separation problems, cut selection, numerical issues, branch-and-cut, interaction between cuts and branching, examples from knapsack, facility location, and network design

Decomposition Methods I (Lagrange):
Motivation for decomposition in large-scale optimization, complicating constraints, Lagrangian relaxation, Lagrangian dual problem, subgradient methods, upper and lower bounds, duality gaps in integer programming, applications to assignment, scheduling, network, and location problems

Decomposition Methods II (Benders, Dantzig-Wolfe):
Block-structured linear and mixed-integer programs, Benders decomposition, master problem and subproblem, feasibility and optimality cuts, Dantzig–Wolfe decomposition, column generation, restricted master problems, pricing problems, branch-and-price, examples from facility location, vehicle routing, and crew scheduling

Stochastic Optimization:
Decision making under uncertainty with probability distributions, recourse decisions, two-stage stochastic linear programs, scenarios, expected-value objective functions, deterministic equivalent formulations, sample average approximation, comparison with robust optimization, applications in logistics, energy, and finance

Part II: Nonlinear Programming and Convex Optimization

Multi-Variable Calculus Review:
Review the mathematical foundations used throughout the course, including gradients, Hessians, continuity, compactness, Lipschitz continuity, and monotonicity.

Nonlinear Optimization Problems: Examples and Motivation:
Introduce the general nonlinear optimization problem and survey problem classes that arise in practice, including least-squares, system identification, minimum cost flows, and optimal power flow. We discuss the distinction between global and local solutions and motivate the role of convexity in making optimization tractable.

Convex Functions and Convex Sets:
Define convex sets and convex functions, survey key examples, and study operations that preserve convexity and relaxations to convexity. First and second order characterizations of convexity are developed, along with the central result that local minima of convex functions are global, connecting geometric intuition to optimality conditions.

TA-Session - CVXPY:
Introduction to CVXPY modeling framework and transcribing optimization problems into code.

Classes of Convex Programs:
The challenges of non-convex problems are described in greater detail, i.e., local optima and saddle points, and relaxations are described. We survey the standard classes of convex optimization problems, including LP, QP, and QCQP, and discuss how a broad range of problems can be transcribed into these forms. Multiobjective optimization is introduced through regularized least squares, scalarization, and Pareto optimality. Additional examples include geometric programs and quasi-convex problems.

Robust Optimization and Conic Programming:
Introduce second-order cone programs (SOCP) and semidefinite programs (SDP) as generalizations of LP and QP. Modeling examples include robust linear programs, norm minimization, and eigenvalue optimization. We discuss applications in control, combinatorics, and signal processing, and methods for transcribing optimization problems into solver-compatible forms.

Duality I:
Develop the theory of duality for convex programs, beginning with conjugate functions and the Lagrangian. Dual variables and the dual problem are derived, and conditions for strong duality are established. We derive KKT optimality conditions and discuss the role of constraint qualifications.

Duality II:
Explore the practical interpretation of duality through penalties, shadow prices, and sensitivity analysis. The implicit function theorem is used to derive solution maps under perturbation, and we develop the envelope theorem and properties of the value function for parametric optimization problems.

TA-Session - Solver Tools:
Compare different solvers for different optimization problems using the CVXPY modeling framework. Examples include SCS, ECOS, MOSEK, GUROBI, and GLPK.

Algorithms & Solvers I:
Develop the core algorithms for smooth constrained and unconstrained optimization. Gradient descent is established as a first-order method to reach optima in unconstrained settings. Projected gradient descent is introduced for constrained problems, motivating the need to handle constraints without resolving a projection at every step. Newton's method is derived, and its local convergence properties are established. Interior point methods via the log barrier are then developed as a practical alternative, connecting barrier methods to duality and KKT conditions.

Algorithms & Solvers II:
Develop algorithms for structured convex problems. The augmented Lagrangian method is introduced as a convergence-improving extension of Lagrangian relaxation. The proximal operator is defined and proximal gradient descent derived, with projected gradient as a special case. ADMM is then developed as a splitting

method that alternates proximal steps across problem blocks, inheriting the convergence properties of the augmented Lagrangian.

Distributed Optimization:
Introduce distributed optimization as a framework for solving large-scale optimization problems across multiple agents connected by a communication network. Topics include consensus optimization, distributed gradient methods, and federated learning, illustrating how decomposition, local computation, and limited information exchange can be used to solve optimization problems without centralized coordination.

Applications: Estimation, Regression, and SVM:
Apply convex optimization to statistical estimation and learning problems. Topics include least-squares and robust regression, regularization, system identification, and support vector machines, illustrating how problem structure can be exploited for efficient solutions.

Applications: Optimal Control:
Apply convex optimization to control problems. Discrete-time open-loop LQR is formulated as a quadratic program, and model predictive control is developed as a receding-horizon extension. We then introduce continuous-time optimal control via the Hamiltonian and Pontryagin's minimum principle.

Application: Network Routing and Population Games:
Introduce population games as a model for large-scale strategic interaction and define Nash equilibrium and potential functions. Specific focus on network congestion games and selfish routing. Population dynamics and their equilibria are discussed, connecting optimization and variational inequality concepts to evolutionary and multi-agent systems.

Assignments and Course Work

Homeworks

The course will contain four homework assignments covering roughly two weeks of the course's content each. The topic and timeline of each homework assignment will roughly follow:

- Homework 1 - Linear Programming - Assigned Week 1, due week 3
- Homework 2 - Mixed Integer Programming - Assigned Week 3, due week 5
- Homework 3 - Convex Optimization - Assigned Week 5, due week 7
- Homework 4 - Duality and Solvers - Assigned Week 7, due week 9

Project

The culminating experience in the course will be a project in which students apply their knowledge of optimization to problems outside the scope of the course lectures. Students can select projects from a list of topics or propose their own. Possible project structures include

1. *Literature based project* - Reproduce the optimization formulation and solution from a published paper, then test a modification to the model or method and analyze the effect.
2. *Problem formulation project* - Encode a real-world problem as an optimization problem, solve it computationally, and critically evaluate the modeling choices and approximations made.
3. *Solver design project* - Implement an optimization solver, test it across problem instances of varying structure, and analyze its behavior, limitations, and tradeoffs relative to problem type.

Successful projects will include an analysis component, a computational component, and a discussion component. Progress and completion of the project will be assessed in three stages: a project proposal, a project report, and a project presentation in class at the end of the term.

- Project Proposal - Due Week 6 - 1-2 page project description and discussion with one or both instructors to ensure sufficient and attainable scope.
- Project Presentation - In class week 10 - 15 minute presentation on project + Q&A from classmates and instructor.
- Project Report - Due Week 10 - report containing analysis, computation, and discussion of project.

AB, BE, and MEng students can complete the project in teams of two. PhD students should complete individual projects.
Grading: Each homework is worth 15% of the final grade. The project is worth 40% (5% proposal, 15% presentation, 20% final report).